

Name: _____

Class: _____

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**CATHOLIC JUNIOR COLLEGE
JC2 PRELIMINARY EXAMINATION**

BIOLOGY**Paper 3 Applications Paper****9747/03****Wednesday 10 September 2008****1 hour 30 min**

Additional materials: Answer booklet
 Writing Paper

READ THESE INSTRUCTIONS FIRST

Write your name and HT group on all the work you hand in.
Write in dark blue or black pen.

You may use a soft pencil for any diagrams, graph or rough working.

Do not use paper clips, highlighters, glue or correction fluid.

Answer **all** questions.

Write your answers in the spaces provided on the question paper.
Show All working

At the end of the examination, fasten all your work securely together;

The number of marks is given in brackets []
at the end of each question or part question.

For Examiner's Use	
1 [15]	
2 [15]	
3 [15]	
4 [20]	
Total	65

This document consists of **8** printed pages.

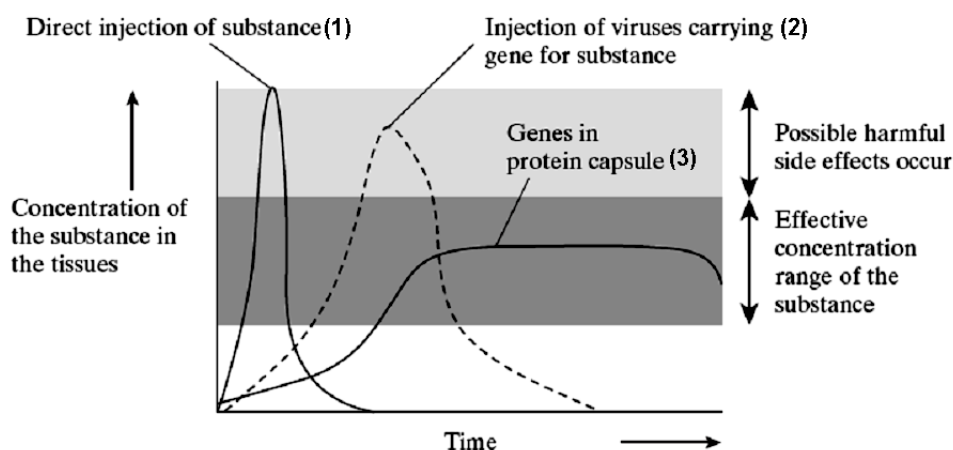
[Turn over]

Answer **all** questions in the spaces provided

- 1 In gene therapy, genes are introduced into a person who has defective genes which do not produce an important substance. Three experiments were done to compare techniques for introducing an important substance into a person with defective genes.

1. The substance was injected directly.
2. Harmless viruses carrying genes coding for the substance were injected.
3. The genes were put into a protein capsule which was inserted into the tissues.

The graph shows results of the experiments.



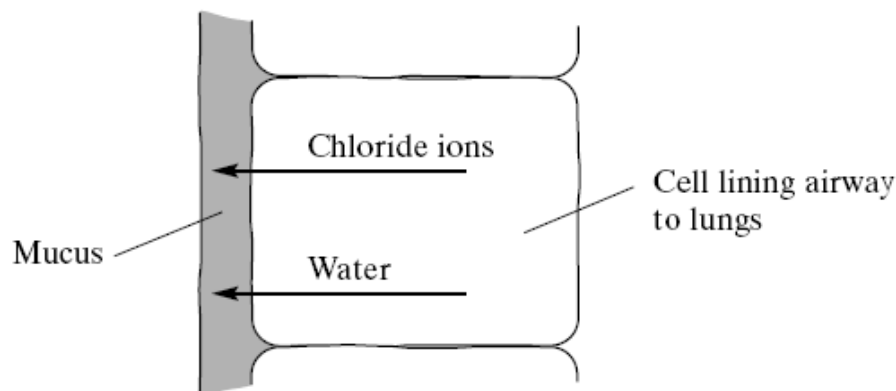
- (a) (i) State three key differences between graphs 2 and 3. [3]

- (ii) Using the information in the graph, suggest **two** advantages and **one** disadvantage of the capsule method compared to the others. [3]

- (b) (i) Some people are concerned about using gene therapy to treat genetic disorders in humans. Give **three** arguments against the treatment of disorders by gene therapy. [3]

- (ii) Explain why a person cured by gene therapy may still have children who suffer from the abnormality. [2]

The diagram shows the movement of chloride ions and water in the cells of the lung of a healthy human.



- (c) (i) In people with cystic fibrosis the gene controlling the movement of chloride ions is defective. The defective gene produces defective CFTR. Suggest why defective CFTR does not transfer chloride ions across cell membranes. [1]

- (ii) People with cystic fibrosis produce mucus which is stickier and thicker than healthy people. Suggest why people with cystic fibrosis are more likely to develop bacterial infections, such as pneumonia. [1]

- (iii) Describe briefly **one** way in which normal CFTR genes might be introduced into the cells lining the airways of the lungs. [2]

[Total : 15]

- 2 Restriction Enzymes (also known as restriction endonucleases) are used to cut DNA molecules.

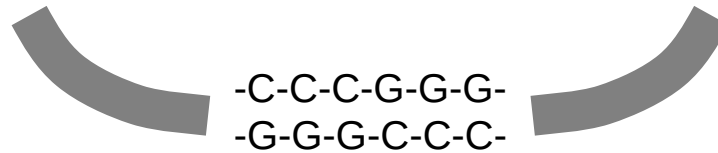
- (a) State the source and natural use of restriction enzymes. [2]

The target sites for three restriction enzymes are shown in Table 2.1.

Table 2.1

Name of restriction enzyme	DNA base sequence of target site
A	<pre> -C-C-C G-G-G- -G-G-G C-C-C- </pre>
B	<pre> -C- C-G-G- -G-G-C- C- </pre>
C	<pre> -A -A-G-G-T-T- -T-T-C-G-A- A- </pre>

Fig 2.1 shows part of a plasmid, including details of the target site for restriction enzyme **A**.



(b) Explain

- (i)** whether it would also be possible to cut this sequence using restriction enzymes **B** and **C**. [2]

enzyme **B**

enzyme **C**

- (ii)** why restriction enzymes will only act at certain target sites within the DNA molecule. [3]

(c) Plasmids are loops of DNA which are found in bacteria. They can be used to transfer genes into bacterial cells. In some cases, these may be human genes.

- (i)** Describe the steps that need to be taken in order to insert a human gene into a plasmid. [4]

- (ii) Using a named example, explain the advantages of being able to transfer human genes into bacteria. [4]

[Total 15]

- 3 Equal masses of tobacco plant callus were cultured for four weeks on media containing different concentrations of two plant growth regulators: auxin and cytokinin. The results are shown in Table 3.1.

Table 3.1

treatment	concentration of plant growth regulators / mg dm ⁻³		effect of plant growth regulators on callus growth
	auxin	cytokinin	
A	2.00	0.00	little or no growth
B	2.00	0.02	growth of roots
C	2.00	0.20	increased growth of callus with no differentiation
D	2.00	0.50	growth of shoots
E	0.00	0.20	little or no growth

- (a) Explain why growing plantlets from a callus in tissue culture results in a clone. [3]

(b) State why the plant tissue cultures are initially grown in sterile conditions. [1]

(c) With reference to Table 3.1,

(i) Describe the effects of auxin and cytokinin on callus growth. [3]

(ii) Suggest what treatment should be given to the shoots developed from callus in treatment **D** to turn them into plantlets. [2]

(d) Outline, using **one** named example, the process of genetic engineering that could increase crop yield. [4]

- (e) Identify **one** possible hazard and benefit of plant genetic engineering. [2]

[Total: 15]

Write your answers on the separate answer paper provided.

Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.

Your answers must be in continuous prose, where appropriate.

Your answers must be set out in sections (a), (b) etc., as indicated in the question.

- 4 (a) Compare the mutations responsible for cystic fibrosis (CF), X-linked and ADA-deficient severe combined immunodeficiency (SCID) and comment on their pattern of inheritance. [10]
- (b) With the recent food shortages world wide this year 2008, there has been a greater demand for alternative sources of food. Discuss some of the ethical and social implications of using genetically modified organisms to solve the demand for food. [10]

[Total: 20]